



Bank Customer Churn Analysis

A Python + Power BI project analysing churn for 10,000 synthetic bank customers.

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Tools: Python (Pandas, NumPy, Matplotlib, Seaborn), Jupyter Notebook, Power BI.

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Abstract & Core Workflow

Scope

End-to-end analysis of 10,000 customers to explain and visualise churn (21 raw + 7 engineered columns).

Python Prep

Load → inspect → clean → validate
→ engineer features (7 new) →
export cleaned CSV.

Power BI

Four-page interactive dashboard: Overview, Churn Drivers, Segmentation, Customer Details.

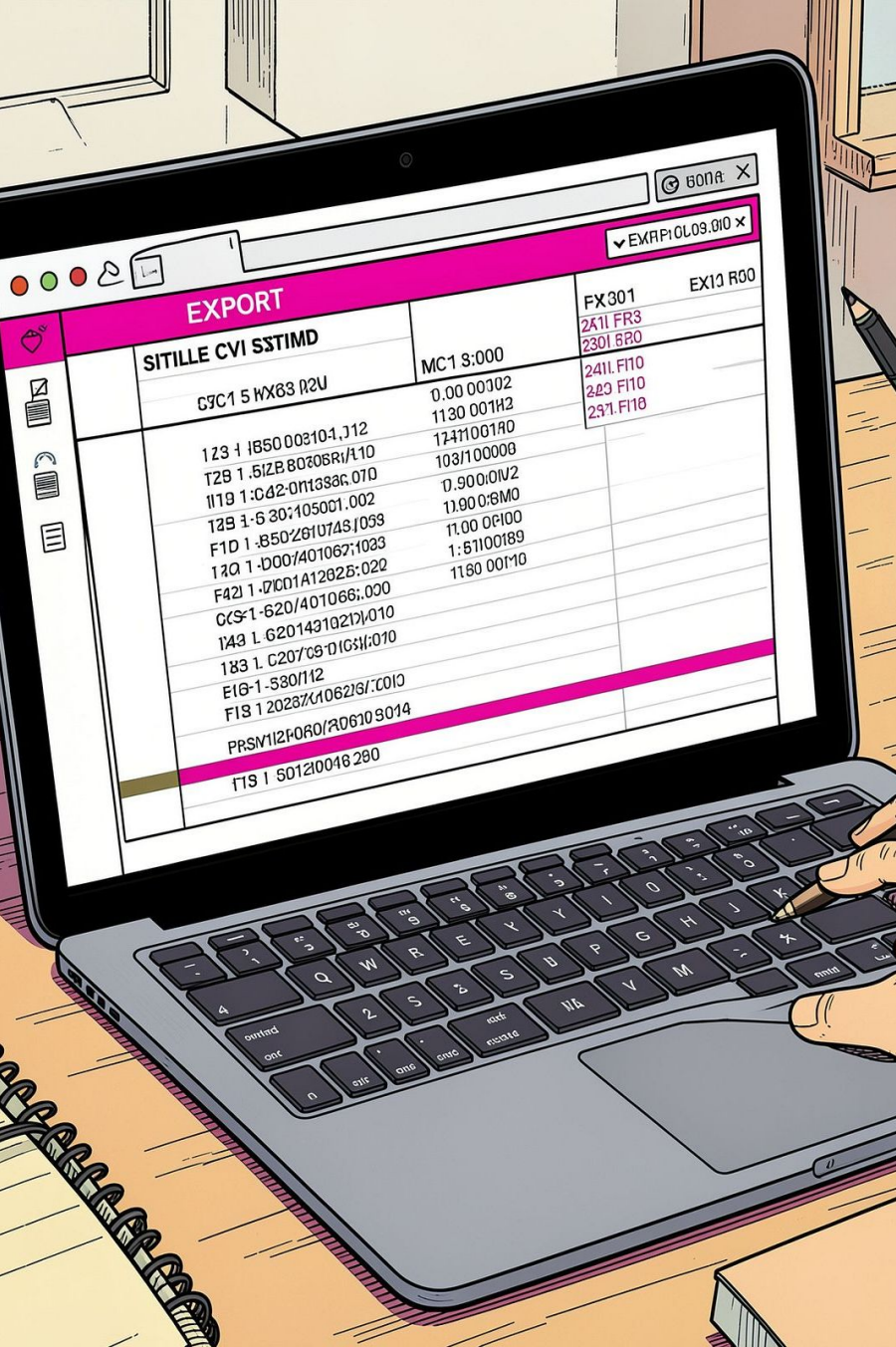
Dataset & Engineered Features

Raw dataset

10,000 rows, 21 columns (Customer_ID, Full_Name, Age, Gender, State, City, Account_Type, Tenure_Years, Account_Balance, Num_Products, Has_Credit_Card, Loan_Status, Credit_Score, Monthly_Income, Loan_Amount, Transactions_Per_Month, Last_Transaction_Date, Mobile_Banking, UPI_Usage, Complaints, Churn).

Engineered columns (7)

- Churn_Status (Stayed/Churned)
- Age_Group (18-25,26-35,36-45,46-55,56+)
- Credit_Score_Group (Poor→Excellent)
- Balance_Category (No Balance→Very High)
- Income_Category (quartiles)
- Days_Since_Last_Transaction
- Activity_Category (Active → Highly Inactive)



Data Quality & Final Export

1

Validation

Range and logical checks: no missing values, no duplicates, all ranges valid (Age 18–75, Credit Score 420–850).

2

Quality Report

Rows: 10,000 ·
Columns: 28 ·
Duplicates: 0 ·
Missing: 0 ·
Churned: 1,838 ·
Churn rate: 18.38%

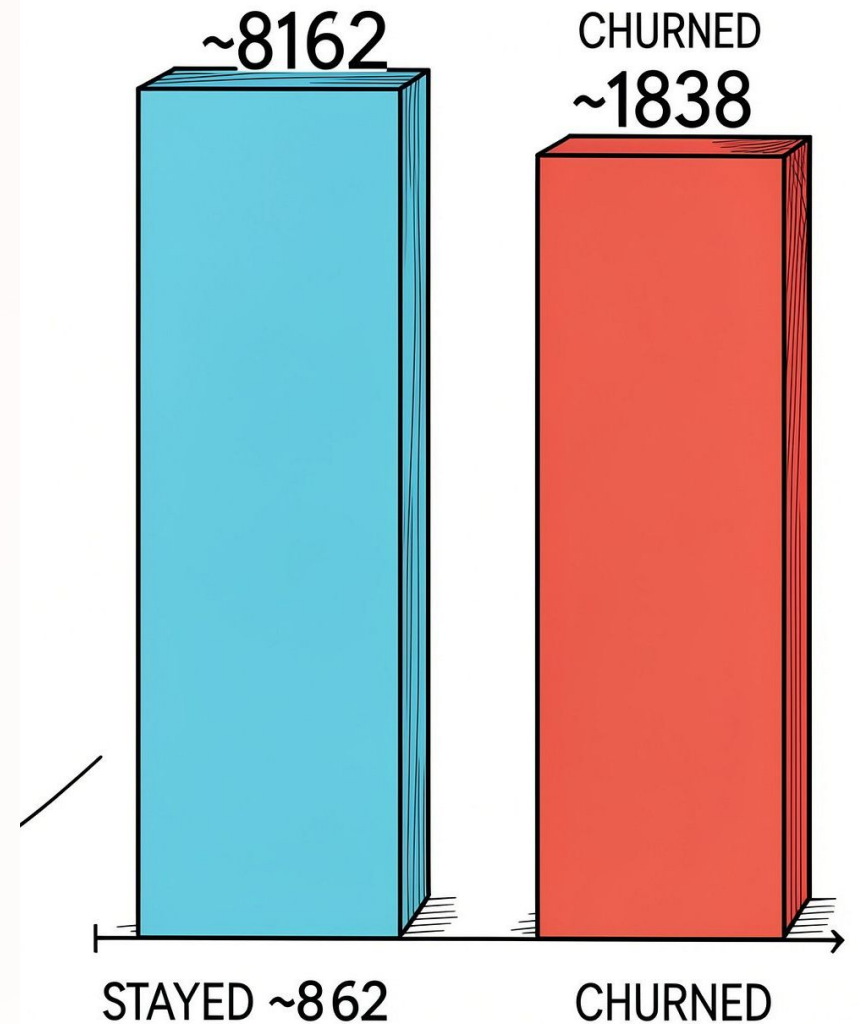
3

Export

Final cleaned file:
bank_customer_churn_cleaned.csv —
used for Power BI.

Overall Churn: Baseline

Out of 10,000 customers: 8,162 stayed (81.62%), 1,838 churned (18.38%). This baseline frames every segment comparison.



Key Univariate Findings



Mobile Banking

Non-users churn 23.22% vs 16.99% for users
— strong protective factor.



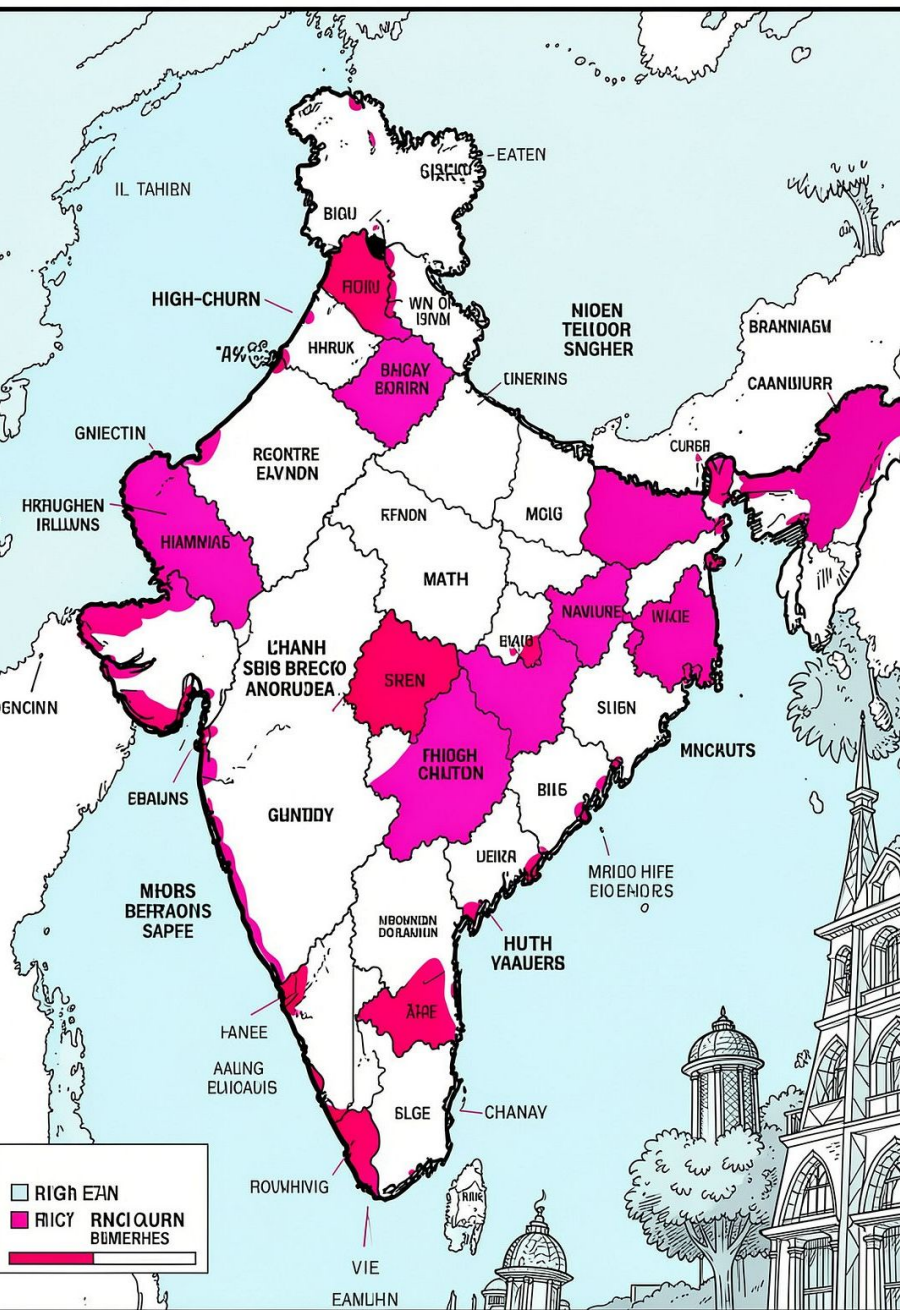
Complaints

Churn rises with complaints: 0→16.31%,
1→18.59%, 2→22.66%, 3→27.33%, 4→56.10%.
Strongest single correlate.



Number of Products

1 product: 20.35% churn → 3 products:
14.86% (protective effect from cross-sell).



Segment Patterns (Demographic & Geographic)

Age & Tenure

Churn increases with age (18–25: 16.03% → 56+: 20.79%). First-year customers highest churn at 21.28% (early-tenure risk).

Geography

State range: Haryana highest 23.66% → Odisha lowest 14.14%. Top churn city: Gurugram 26.56% (city-level targeting advised).



Account & Financial Signals



Account Type

Premium churn 19.18% (highest), Salary accounts most stable 17.22% — prioritize high-value outreach.



Credit Score & Balance

Credit-score and balance show flat effects (roughly 17–20% across bands). Alone, weak predictors.



Loan Status

No Loan customers churn 20.07% vs Active loan 17.28% — lending relationships increase stickiness.

Multivariate View & Dashboard

Correlation insight

Pearson correlations with Churn are weak: all coefficients between -0.05 and +0.08. Complaints (+0.08) and Num_Products (-0.05) lead the list.

Power BI dashboard

Four pages — Overview (KPIs & breakdowns), Churn Drivers (behaviour & service signals), Segmentation (portfolio metrics), Customer Details (searchable profile with churn score-ready fields).



Business Recommendations & Next Steps



Complaint Recovery

Route customers with 2+ complaints into proactive recovery workflows; escalate before 3–4 complaints.



Cross-sell Strategy

Prioritise 1→2 and 2→3 product cross-sell, especially within first 12 months.



Digital Adoption

Drive mobile banking and UPI activation campaigns for non-digital customers to reduce churn.



Targeted Regional Action

Focus retention resources on high-churn states/cities (e.g., Haryana, Gurugram) with local root-cause analysis.



Future Work

Build predictive models (Logistic/Tree/GBM), use SHAP for feature importance, add CLV, cohort/time-series analysis, and automate data refresh into Power BI.



Overall takeaway: combine complaint monitoring, product depth, digital adoption, and early-tenure care into a coordinated retention program supported by the Power BI dashboard.